

Priyanshi Agrawal

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Personal

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Academic Credentials

2024–	PhD in Computer Science (Ongoing) (CPI: 9.17) Indian Institute of Technology Gandhinagar
2022–2024	M.Sc. in Mathematics (CPI: 7.51) Indian Institute of Technology Gandhinagar
2019–2022	B.Sc. in Mathematics (Outstanding: 8.36) St. Xavier's College, Ahmedabad

Current Research

Working on combinatorial algorithms for induced subgraph detection in graphs. Current focus includes polynomial-based methods, exploiting automorphism parity, and designing efficient algorithms for detecting small induced patterns such as paths and cycles.

Master's Thesis

2023–2024	Mathematical Structures in Data and Music During the first semester of my master's thesis, I worked on Topological Data Analysis, where I studied how single linkage clustering can be used to cluster data points in high dimensions. I implemented clustering algorithms in Python and applied them to institutional datasets. During the second semester, I explored the relationship between Mathematics and Music. I studied algorithmic methods to compute rank and metrical patterns of Swara permutations in Rev-Colex order and implemented several of these algorithms in Python.
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Academic Visits and Conferences

2024	Frontiers of Geometric Algorithms, IISc Bangalore
2025	Frontiers of Graph Algorithms, IISc Bangalore
2024	Foundations of Software Technology and Theoretical Computer Science (FSTTCS), IIT Gandhinagar
2026	Foundations of Software Technology and Theoretical Computer Science (FSTTCS), BITS Pilani, Goa Campus

Academic Achievements

2017	International Informatics Olympiad: World Rank 47, State Rank 1.
2022	Secured admission among 500 applicants to the Early Admit M.Sc. Program , IIT Gandhinagar.
2022	Certified in A.R. Rao Mathematics Competition.
2022	Zonal Topper, Mimamsa, IISER Pune.
2020, 2022	Gold Medalist, Insight Coding Competition, L.D. Engineering College.

Cultural Achievements

2023–2025	Winner, Annual Poetry Competition <i>Unkahi</i> , IIT Gandhinagar (organized by BlithChron).
2024	Wrote and directed the winning Street Play, Inter IIT Cultural Meet 7.0.
2023	Top 16 Poets of India, Kshitij Poetry Foundation.
2025	Winner, Poetry Competition, DA-IICT Cultural Meet.

Positions of Responsibility

2025–26	Secretary, Literary Club, IIT Gandhinagar.
2025–26	Co-organiser, CS Theory Seminar series
2024	Content Analyst, TEDxIITGandhinagar.

Leadership, Service, and Sports

2021–2022	National Cadet Corps (NCC): Awarded ‘A’ Grade in B and C certificates; held the rank of Leading Cadet and trained junior cadets during CATC(Combined Annual Training Camp).
2019–2022	National Service Scheme (NSS): Volunteered in multiple social service and community outreach activities.
2014–2015	Bronze Medalist (Twice), State Wushu Championship.
2014–2015	Gold Medalist (Twice), District Wushu Championship.